SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1515

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I***KESHAVARAM L/192511253*** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

Assessment Tasks

Analytical Problem Solving

Q1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Answer:

Criteria	**Type-1 Hypervisor**	**Type-2 Hypervisor**
Definition	Runs directly on physical hardware.	Runs on top of a host operating system.
Performance	High performance with low overhead.	Lower performance due to extra overhead.
Security	Smaller software layer and generally smaller attack surface.	Security depends on the host operating system.
Manageability	Supports centralized management of servers and VMs.	Easier to use for desktop testing.
Usage	Suitable for large cloud datacenters and mission-critical applications.	Suitable for personal use, testing, and development.
Examples	VMware ESXi, Microsoft Hyper-V, Xen.	Oracle VirtualBox, VMware Workstation.
Conclusion	**Best choice** for cloud datacenters because of performance, scalability, and centralized management.	Less suitable for large, mission-critical cloud environments.

Q2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Answer:

Virtualization uses a hypervisor to create separate virtual machines with isolated operating systems, memory, CPU allocations, virtual disks, and virtual networks. Role-based access control, network segmentation, encryption, and resource quotas add protection.

Two major attack surfaces are:

1. Hypervisor vulnerabilities: an attacker may exploit a flaw to escape a VM and access the host or other VMs.

2. Virtual network or management interfaces: weak credentials, exposed APIs, or misconfigured virtual networks may allow unauthorized access.

The provider should patch the hypervisor, restrict management access, use multi-factor authentication, separate sensitive networks, monitor logs, and test isolation regularly.

Q3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Answer:

CPU at 92% indicates high processor demand and possible CPU contention. Memory at 88% leaves little headroom and may cause swapping or ballooning. Storage I/O latency of 35 ms suggests slow or overloaded storage, which can delay disk operations and VM startup. An average boot time of 170 seconds may result from CPU, memory, storage, or startup-service bottlenecks.

Corrective actions:

1. Move some VMs to less-loaded hosts and balance workloads.

2. Add CPU or memory resources where necessary and right-size oversized VMs.

3. Use faster storage, check disk queues, and distribute I/O-heavy workloads.

4. Review VM startup services and boot configuration.

5. Monitor resource use and set alerts before utilization reaches critical levels.

Q4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Answer:

A Software Defined Datacenter (SDDC) uses software to control and automate compute, storage, and networking resources. In a traditional datacenter, many tasks require manual configuration of separate hardware systems.

Compute virtualization: VMs can be created, resized, migrated, and managed through software.

Storage virtualization: storage from multiple devices can be pooled and allocated as needed, improving flexibility and utilization.

Network virtualization: virtual networks, software-defined policies, and automated configuration make it easier to isolate applications and apply security rules.

SDDC improves agility by enabling faster provisioning, consistent policies, automation, scaling, and simpler disaster recovery. It can reduce manual work, though it requires careful security, monitoring, and management.

Q5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Answer:

Identity mismatches may occur because providers use different user identifiers, roles, authentication methods, or token formats. These differences can cause login failures or grant users excessive permissions.

A secure design should:

1. Use a trusted identity provider and federation standards such as SAML 2.0 or OpenID Connect.
2. Map roles and attributes consistently between providers and apply least-privilege access.
3. Use multi-factor authentication and short-lived, securely signed tokens.
4. Encrypt data in transit with TLS and sensitive data at rest.
5. Avoid sharing unnecessary personal information; use only required identity attributes.
6. Centralize audit logs, monitor suspicious logins, rotate keys, and review access regularly.

This approach provides consistent authentication, controlled authorization, and better privacy across cloud providers.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: